

Updated functional requirements:

Functional Requirements — v2

Each requirement traces to a stage of the core user journey: Registration → KYC → Beneficiary → Quote → Cash-in → Queue/Settlement → Wallet → Cash-out, plus supporting Admin functions.

3.1 Registration & Authentication

ID	Requirement
FR-01	The system shall allow a new user to register an account with a full name, email address, mobile number, and password.
FR-02	The system shall allow a registered user to log in using their registered credentials.
FR-02a [NEW]	The system shall allow a logged-in user to log out, terminating their session.
FR-03	The system shall enforce that the email address and mobile number are unique across registered accounts.
FR-04	The system shall distinguish between sender, recipient, and administrator roles.
FR-04a [NEW]	The system shall allow a logged-in user to view and update their basic profile information (e.g. full name, email, mobile number).

3.2 KYC

ID	Requirement
FR-05	The system shall collect the following KYC information from a user: full name, date of birth, nationality, identification number, residential address, mobile number, email address, and source of funds.
FR-06	The system shall allow a user to submit a completed KYC application for review.

FR-07	The system shall allow an administrator to view a submitted KYC application.
FR-08	The system shall allow an administrator to approve or reject a submitted KYC application.
FR-08a [NEW]	The system shall allow a user to view their own current KYC status (e.g. not submitted, pending, approved, rejected).
FR-09	The system shall prevent a user without an approved KYC application from sending a remittance.
FR-09a [NEW / UPDATED ASSUMPTION]	The system shall prevent a recipient without an approved KYC application from requesting a cash-out.

Assumption: Both senders and recipients must have an approved KYC application before they can move value out of the platform - senders before initiating a remittance (FR-09), recipients before cashing out RLUSD to fiat (FR-09a). A recipient may still receive and hold RLUSD in their custodial wallet without KYC, but cannot cash out until approved. This should be stated explicitly in the "Assumptions and Limitations" section of the technical specification.

3.3 Beneficiary Management

ID	Requirement
FR-10	The system shall allow an approved sender to add a beneficiary record containing: full name, mobile number or email address, country, preferred payout currency, and relationship to the sender.
FR-11	The system shall require at least a mobile number or an email address when a beneficiary is added.
FR-12	The system shall allow a sender to view the list of beneficiaries they have added.
FR-12a [NEW]	The system shall link a beneficiary record to a recipient user account by matching the beneficiary's mobile number or email address against an existing registered account.
FR-12b [NEW]	The system shall only provision a custodial RLUSD wallet for a beneficiary once they are linked to a registered, authenticated recipient account.

FR-12c
[NEW] If no registered account matches the beneficiary's mobile number or email address, the system shall leave the beneficiary record unlinked and shall not provision a wallet or allow settlement until a matching account is registered.

Assumption: we need to decide on a chosen linking mechanism (auto-match on existing account vs. invite-to-register flow) in the technical specification - this is currently unspecified in the brief.

3.4 Quote, Fees & Limits

ID	Requirement
FR-13	The system shall determine the applicable USD/ZAR exchange rate at the moment a remittance transaction is created, using a public API, mock service, or configured rate table.
FR-14	The system shall generate a quotation showing: ZAR send amount, exchange rate used, transaction fee, foreign-exchange margin, RLUSD amount to be received, cash-out fee, and estimated recipient payout.
FR-15	The system shall calculate fees using configurable parameters: a fixed fee, a percentage-based fee, an FX margin, and a cash-out fee.
FR-15a [NEW]	The system shall provide an administrator interface for viewing and updating the configurable fee parameters (fixed fee, percentage fee, FX margin, cash-out fee).
FR-16	The system shall check a requested remittance amount against the sender's configured daily and monthly limits before allowing the transaction to proceed.
FR-16a [NEW]	The system shall allow a sender to view their current daily and monthly remittance limits and their remaining available allowance.

- FR-16b The system shall provide an administrator interface for viewing and updating
[NEW] the daily and monthly limit tiers (e.g. unverified vs. verified user levels).
- FR-17 The system shall reject a remittance request that would cause the sender to
exceed either their daily or monthly limit, and shall inform the sender of the
reason.

3.5 Simulated Cash-In

- | ID | Requirement |
|-----------|---|
| FR-18 | The system shall allow a sender to initiate a simulated ZAR payment via one supported method (e.g. agent cash, bank transfer, or card). |
| FR-19 | The system shall allow an administrator or mock payment service to confirm that a simulated ZAR payment has been received. |
| FR-20 | The system shall not initiate RLUSD settlement for a transaction until the corresponding ZAR cash-in has been confirmed. |

3.6 Queue & Settlement

- | ID | Requirement |
|-----------|---|
| FR-21 | The system shall place a settlement message on a message queue once a ZAR cash-in is confirmed. |
| FR-22 | A settlement worker shall consume queued settlement messages and submit the corresponding RLUSD transfer to the XRP Ledger Testnet. |
| FR-23 | The system shall record the outcome (success or failure) of each submitted XRPL transaction, including the transaction hash on success. |
| FR-24 | The system shall handle a failed XRPL transaction without crediting the recipient's wallet. |
| FR-25 | The system shall ensure that a duplicate settlement message cannot credit a recipient's wallet more than once for the same transaction. |
| FR-26 | The system shall update the recipient's wallet balance once RLUSD settlement is confirmed on-ledger. |

3.7 Recipient Wallet & Sender Transaction History

ID	Requirement
FR-27	The system shall provide each recipient with a custodial web wallet.
FR-28	The wallet shall display: available RLUSD balance, incoming transfers, outgoing/cash-out transactions, transaction status, transaction date, and XRPL transaction hash.
FR-28a [NEW]	The system shall allow a sender to view the status and history of remittances they have sent, including quote details, cash-in status, settlement status, and (where applicable) the XRPL transaction hash.

3.8 Simulated Cash-Out

ID	Requirement
FR-29	The system shall establish the necessary TrustLine (TrustSet) to the RLUSD issuer for any wallet that is to hold RLUSD.
FR-30	The system shall allow a recipient to request a cash-out from RLUSD to USD or a supported local fiat currency.
FR-31	The system shall calculate the fiat payout amount using the applicable exchange rate, less the configured cash-out fee.
FR-32	The system shall track each cash-out request through the states: requested, approved, completed, or failed.

3.9 Administration

ID	Requirement
FR-33	The system shall provide an administrator interface for reviewing and actioning KYC applications.
FR-34	The system shall provide an administrator interface for confirming simulated cash-in payments.
FR-35	The system shall provide an administrator interface for actioning simulated cash-out requests.

(FR-15a and FR-16b above also fall under Administration — grouped with Fees/Limits for traceability to the requirement they configure.)

4. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	API endpoints shall respond within 2 seconds under normal (non-load-test) conditions.
NFR-02	Concurrency	The system shall remain functional and responsive under a defined concurrent-user load during performance testing (e.g. 50 simulated concurrent users).
NFR-03	Security - Authentication	User passwords shall be stored using a salted, industry-standard hashing algorithm, never in plaintext.
NFR-04	Security - Private Keys	XRPL private keys shall be encrypted at rest, with the encryption key stored separately from the database holding the encrypted keys.
NFR-05	Security - Private Keys	Private keys shall never be returned via the API, never appear in application logs, never be committed to source control, and shall only be decrypted by the component responsible for signing XRPL transactions.
NFR-06	Security - Logging	Application logs shall exclude sensitive data (private keys, full KYC identification numbers, passwords).
NFR-07	Availability	The system shall remain stable and available for the full duration of the live demo session, with no unhandled crashes.
NFR-08	Data Protection	Customer KYC and personal information shall be access-controlled, viewable only by the account owner and authorised administrators.
NFR-08a [NEW]	Data Protection	Sensitive KYC fields (e.g. identification number) shall be encrypted at rest in the database, in addition to being access-controlled.
NFR-09	Auditability	Every remittance transaction (quote, cash-in confirmation, settlement outcome, cash-out) shall be timestamped and traceable end-to-end.

Changelog from v1

- **Added:** FR-02a (logout), FR-04a (view/edit profile), FR-08a (view own KYC status), FR-16a (view own limits/allowance) - manage basic profile information / log in and log out / view KYC status / view transaction limits" requirements.
- **Added:** FR-09a and the KYC-before-cash-out assumption - recipients now require approved KYC before cash-out, mirroring FR-09 for senders.
- **Added:** FR-12a, FR-12b - recipient onboarding/account-linking is now an explicit requirement rather than implied; flags a design decision your team needs to make and document.
- **Added:** FR-15a, FR-16b - admin interfaces for configuring fees and limit tiers (previously "configurable" was stated but no admin-facing FR existed).
- **Added:** FR-28a - sender-side transaction history, distinct from the recipient wallet view in FR-28.
- **Added:** NFR-08a - encryption at rest for sensitive KYC fields, not just access control.
- **Unchanged:** FR-13–FR-27, FR-29–FR-35, NFR-01–NFR-07, NFR-09 (matched brief)